PEMBAHASAN SOAL UTS

1. Host A receives a frame and discards it after determining it is corrupt. Which OSI layer checks frames for errors?

0/15 A Application
0/15 B Network
0/15 C Physical
15/15 D Data-Link

2. Which of the following is not an MAC address type?

1/17 A Unicast
2/17 B Broadcast
13/17 C Anycast
1/17 D Multicast

Unicast, Broadcast, Multicast

Mac address & IPv4

Anycast => Ipv6

unicast
Multicast

3. Which of the following is correct about Half-Duplex Communication?

1/15 A Communication is unidirectional
1/15 B Only 1 node can transmit, the other node can only receive.
11/15 C Both nodes can send and received, but not at the same time.
2/15 D Both nodes can send and received simultaneously

4. Which layer of the OSI model is responsible for reliable delivery of data across the physical network?

1/12 A Network Layer
1/12 B Data Link Layer
10/12 C Transport Layer
0/12 D Physical Layer

Wired => Ethernet, Fiber,
Wireless => Radio, Wifi, Bt

transport
network
data-link
phy

5. Which layer of the OSI model is responsible for managing sessions between applications?

0/13 A Presentation Layer
0/13 B Application Layer
13/13 C Session Layer
0/13 D Transport Layer

user => information

6. Which of the following are not steps in the data-encapsulation process?

6/19 **A** Segments are converted into frames

5/19 **B** Frames are converted into bits

2/19 **C** Packets are converted into frames

6/19 **D** User information is converted into data

upper layer (app, pres, sessi) => data
Transport => Segment
Network => Packet
Data-link => Fram
Pyh => bits

7. Which layer of the OSI model theoretically resides closest to the end user?

12/15 **A** Application Layer

1/15 **B** Presentation Layer

0/15 **C** Physical Layer

2/15 **D** Data-Link Layer

8. Computer networks do which of the following?

10/19 **A** Allow computer hosts to communicate data between each other

2/19 **B** Provide a user interface to control computer hosts

7/19 **C** Provide a user interface to control networking devices

0/19 **D** Operate solar power stations

9. Networking devices do which of the following?

1/14 **A** Provide an operating system for the Internet

3/14 **B** Allow users to use more than one computer at once

7/14 **C** Control and optimize communication between host devices

3/14 **D** Allow users to use more than one user interface at once

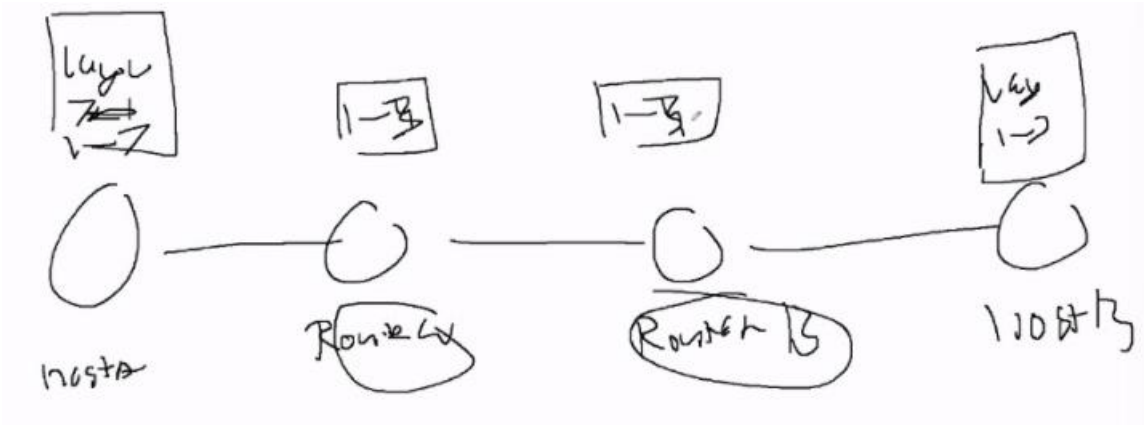
10. What is the Open Systems Interconnection (OSI) reference model?

0/17 **A** A cabling standard that phone and networking companies implement

1/17 **B** A computer host architecture blueprint that computer device manufacturers implement

10/17 **C** A network architecture blueprint that network and computer device manufacturers implement

6/17 **D** A network device architecture blueprint that network device manufacturers implement



11. How does the OSI reference model guarantee independence of layer functionality?

10/21 **A** By encapsulating data at each layer and by defining clear interfaces between each layer

8/21 **B** By allowing network and computer device manufacturers to implement proprietary layers

3/21 **C** By allowing communication between computer hosts over a network

0/21 **D** By optimizing the network route between computer hosts

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12. What is the function of the application layer of the OSI reference model?

0/11 **A** Segments and delivers data, correcting transmission errors transport

4/11 **B** Opens and maintains communication channels, authenticating data communications session

2/11 **C** Converts data, encrypts data, and translates data presentation

5/11 **D** Manages the user interface and processes data before handing it off to the user apps

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User gak bisa langsung masuk ke layer tanpa lewat aplikasi

13. What is the function of the presentation layer of the OSI reference model?

1/13 **A** Segments and delivers data, correcting transmission errors

2/13 **B** Opens and maintains communication channels, authenticating data communications compressi data

8/13 **C** Converts data, encrypts data, and translates data presentation

2/13 **D** Manages the user interface and processes data before handing it off to the user

compresi di presentation

14. What is the function of the session layer of the OSI reference model?

1/20 **A** Segments and delivers data, correcting transmission errors

17/20 **B** Opens and maintains communication channels, authenticating data communications

2/20 **C** Converts data, encrypts data, and translates data

0/20 **D** Manages the user interface and processes data before handing it off to the user

15. What is the function of the transport layer of the OSI reference model?

8/12 **A** Segments and delivers data, correcting transmission errors

0/12 **B** Opens and maintains communication channels, authenticating data communications

2/12 **C** Converts data, encrypts data, and translates data

2/12 **D** Manages the user interface and processes data before handing it off to the user

16. What is the function of the network layer of the OSI reference model?

5/21 **A** Segments and delivers data, correcting transmission errors transport

0/21 **B** Manages electrical, optical, and over-the-air transmission of data bits fisik pyhisical

5/21 **C** Routes data frames locally and manages local physical addressing data link

11/21 **D** Routes data packets between networks and manages global logical addressing network

17. What is the function of the data link layer of the OSI reference model?

9/14 **A** Segments and delivers data, correcting transmission errors

0/14 **B** Manages electrical, optical, and over-the-air transmission of data bits

4/14 **C** Routes data frames locally and manages local physical addressing

1/14 **D** Routes data packets between networks and manages global logical addressing

18. What is the function of the physical layer of the OSI reference model?

0/17 A Segments and delivers data, correcting transmission errors

11/17 B Manages electrical, optical, and over-the-air transmission of data bits

2/17 C Routes data frames locally and manages local physical addressing

4/17 D Routes data packets between networks and manages global logical addressing

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19. What do Media Access Control (MAC) addresses uniquely identify?

0/13 A A specific wide-area network (WAN)

0/13 B A specific local-area network (LAN)

13/13 C A specific network device, such as a switch or a router, or a network interface card (NIC) in a computer host device

0/13 D All of the above

20. Name one of the main functions of Layer 3 (network layer) in TCP/IP protocols.

1/16 A Manage the electrical aspect of network links

9/16 B Choose the best route to send data packets between hosts, even when the hosts are separated by several networks

1/16 C Choose the best route to send data frames between hosts, only when the hosts are located within the same local network

5/16 D All of the above

Yang A layer 1, yang C layer 2 / data link

21. Describe protocol data units (PDUs).

3/13 A Basic encryption method that allows a network interface card (NIC) to send encrypted data packets over the network

4/13 B Basic encoding method that allows a network interface card (NIC) to send data packets over the network

5/13 C Basic data container used by each protocol to exchange data between hosts in a network

1/13 D Basic data container used by each protocol to set up a virtual circuit

22. Each protocol data unit (PDU) contains which of the following?

4/18 A Data payload and control information about the local computer host

8/18 B Data payload and control information that helps the protocol figure out what to do with the data payload

6/18 C Data payload and control information about devices in the network

0/18 D Data payload and control information about the remote computer host

23. On the sending host, what is the function of each network layer?

0/15	A	Receives a PDU from the upper layer
2/15	B	Builds its own PDU containing the upper-layer PDU encapsulated within its own PDU
6/15	C	Encapsulates original data payload in several nested PDUs, one for each layer
7/15	D	All of the above

24. On the receiving host, what is the function of each network layer?

8/19	A	Receives a PDU from the lower layer
1/19	B	Strips out the lower-layer header and trailer because it doesn't need them
6/19	C	Extracts its own PDU, containing a control header, data payload, and optionally a control trailer
4/19	D	All of the above

Header-headernya dilepas untuk mengekstrak pdu nya sendiri

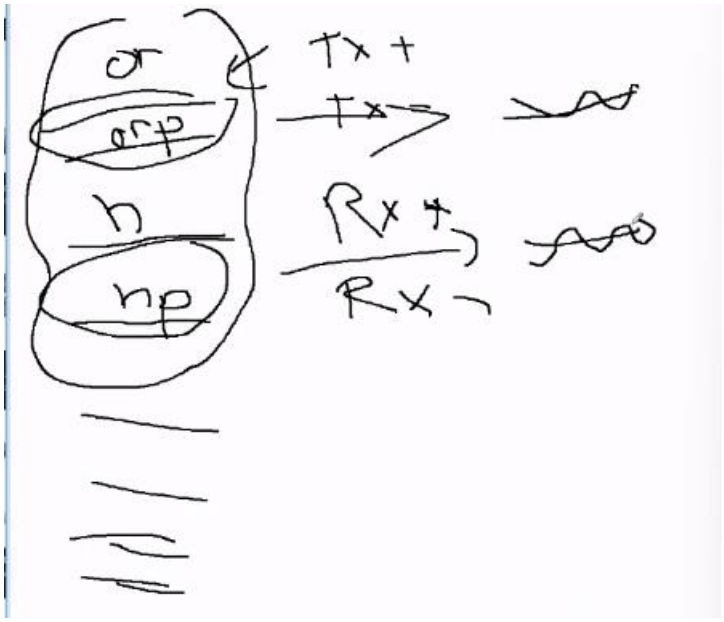
25. Which of the following describes Ethernet?

2/17	A	A standard UTP implementation that specifies characteristics of UTP cabling operation at the physical and data link layers
1/17	B	A standard WAN implementation that specifies characteristics of WAN operation at the physical and data link layers
1/17	C	A standard fiber-optic implementation that specifies characteristics of fiber-optic cabling operation at the physical and data link layers
13/17	D	A standard LAN implementation that specifies characteristics of LAN operation at the physical and data link layers

Ethernet → layer 2 → LAN

26. Which of the following describes full-duplex transmissions?

2/20	A	Data frames are transmitted in one direction only over a single pair of wires
8/20	B	Data frames are transmitted in both directions, simultaneously, over a single pair of wires
7/20	C	Data frames are transmitted in both directions, simultaneously, over two pairs of wires
3/20	D	Data frames are transmitted in both directions, simultaneously, over two wires



UTP dan STP => Twisted Pair

Kalau Single → Half two pairs

27. What is the maximum bandwidth of Ethernet?

2/16 A 10000 Mbps

2/16 B 10 Mbps

9/16 C 1000 Mbps

3/16 D 100 Mbps

Ethernet => 10 Mbps
Fast Ethernet => 100 Mbps
Giga Ethernet => 1000 Mbps => 1 Gbps
Ten Gig Etrhent => 10 Gbps => 10.000 Mbps
hundred gig ethernet => 100 Gbps => 100.000 Mbps

28. Describe wide-area networks (WANs).

3/12 A They span long distances.

1/12 B They interconnect local-area networks (LANs).

0/12 C They interconnect all the networks that make up the Internet.

8/12 D All of the above.

29. What is one of the layers defined by the Hierarchical network model?

3/18 A Data link layer

0/18 B Session layer

14/18 C Access layer

1/18 D Presentation layer

Core Layer
Distribution Layer
Access Layer

30. Which layer of the OSI model contains the MAC and LLC sublayers?

0/14 A Physical layer
0/14 B Network layer
0/14 C Transport layer
14/14 D Data link layer

31. Which of the following is a valid MAC address?

0/14 A 192.168.101.1
0/14 B 0100111011100100
11/14 C 00-0f-1f-9c-2d-ad
3/14 D 255.255.255.0

Klo mac Address : 48 BIT, NOTASI HEKSA

aa-bb-cc-dd-ee-ff

aa:bb:cc:dd:ee:ff

aabb.ccdd.eeff

32. Data framing occurs at which OSI layer?

0/11 A Transport layer
0/11 B Network layer
11/11 C Data link layer
0/11 D Session layer

33. What is the correct order of data encapsulation starting from the application layer to the physical layer?

0/12 A Bits, Frames, Packets, Segments, User Data
0/12 B User data, Frames, Packets, Segments, Bits
1/12 C User data, Packets, Segments, Frames, Bits
11/12 D User data, Segments, Packets, Frames, Bits

34. ARP determines the mac address by using the known ip address.

4/14 A Hardware, IP
0/14 B MAC, HEX
10/14 C IP, MAC RARP => Reverse ARP
0/14 D Ethernet, RARP

35. Hosts use the _____ to determine the MAC address of a target device.

0/18 A UDP (User Datagram Protocol)

13/18 B ARP (Address Resolution Protocol)

1/18 C DHCP (Dynamic Host Configuration Protocol)

4/18 D TCP (Transport Control Protocol)

36. Hosts need to broadcast _____ whenever their _____ does not contain the MAC address of a target device.

13/17 A an ARP request, ARP table

1/17 B a UDP request, UDP table

2/17 C a DHCP request, DHCP table

1/17 D a TCP request, TCP table

Udp layer 4 (gak ada tabel)

Dhcp layer aplikasi

Tcp layer 4 (gak ada tabel)

37. When sending data frames to a host within the LAN, the destination MAC address in the data frame is set to what?

4/15 A The IP address of the destination host

2/15 B The IP address of the LAN gateway

8/15 C The MAC address of the destination host

1/15 D The MAC address of the LAN gateway

source add | dest add

Sama lan destination

Klo beda lan destinationnya berarti gateway

38. When sending data frames to a host in a remote network, the destination MAC address in the data frame is set to what?

0/11 A The IP address of the destination host

2/11 B The IP address of the LAN gateway

5/11 C The MAC address of the destination host

4/11 D The MAC address of the LAN gateway

39. Unicast transmission involves which of the following?

1/12 A A device sending a frame to multiple target devices

0/12 B A device sending a frame to all devices in its local network

8/12 C A device sending a frame to a single target device

3/12 D All of the above

Klo all device broadcast

Select group berarti multicast

40. Describe how a Layer 2 switch learns addresses.

2/15 A The switch saves the port number where each data frame enters the switch along with the source IP address of that frame in the IP address table.

12/15 B The switch saves the port number where each data frame enters the switch along with the source MAC address of that frame in the MAC address table.

1/15 C The switch saves the port number where each data frame enters the switch along with the source firmware version of that frame in the MAC address table.

0/15 D The switch saves the port number where each data frame enters the switch along with the source firmware version of that frame in the IP address table.

41. What is the main function of the data link layer?

18/25 A To handle data frame transmission locally between two devices connected on a local-area network (LAN)

4/25 B To handle data frame transmission between two devices connected on a wide-area network (WAN)

2/25 C To handle IP packet routing locally between two nodes connected on a local-area network (LAN)

1/25 D To handle IP packet routing between two nodes connected on a wide-area network (WAN)

Ip packet mah network

42. What method can be used by two computers to ensure that packets are not dropped because too much data is being sent too quickly?

0/13 A Encapsulation

12/13 B Flow control

0/13 C Framming

1/13 D Error control

43. Which name is assigned to the transport layer PDU?

0/14 A Bits

2/14 B Packets

0/14 C Frame

12/14 D Segment

44. What is a characteristic of multicast messages?

13/13 A They are sent to a select group of hosts.

0/13 B They are sent to all hosts on a network.

0/13 C They are sent to all hosts on a network.

0/13 D They are sent to a single destination

45. Which one is correct about client-server network type?

2/13 A Each computer is responsible for making its own resources available to other computers on the network.

0/13 B Each computer is responsible for setting up and maintaining its own security for these resources.

3/13 C Useful for a small network containing less than 10 computers on a single LAN .

8/13 D Defined by the presence of servers on a network that provide security and administration of the network.

A B C → PEAR TO PEAR

46. Refer to the graphic. What type of cabling is shown?



0/12 A Serial RS-232

12/12 B Ethernet - UTP

0/12 C Coaxial 10Base2

0/12 D Fiber Optic

47. A network team is comparing physical WAN topologies for connecting remote sites to a headquarters building. Which topology provides high availability and connects some, but not all, remote sites?

0/7 A Mesh Topology

7/7 B Partial Mesh Topology

0/7 C Hub and Spoke Topology

0/7 D Point-to-point Topology

48. Which network topology has the highest redundancy that can be built in?

1/11 A Star Topology

0/11 B Bus Topology

0/11 C Ring Topology

10/11 D Mesh Topology

49. At which layer of the OSI model would a logical address be added during encapsulation?

0/11 A Physical Layer

0/11 B Data-Link Layer

10/11 C Network Layer

1/11 D Transport Layer

50. What is an advantage of network devices using open standard protocols?

0/11 A Network communications is confined to data transfers between devices from the same vendor.

11/11 B A client host and a server running different operating systems can successfully exchange data.

0/11 C Internet access can be controlled by a single ISP in each market

0/11 D Competition and innovation are limited to specific types of products.

Osi layer Open standard

51. Which statement is correct regarding the operation of DHCP?

0/14 A A DHCP client uses a ping to detect address conflicts

0/14 B A DHCP server uses a gratuitous ARP to detect DHCP clients.

12/14 C If an address conflict is detected, the address is removed from the pool and an administrator must resolve the conflict.

2/14 D If an address conflict is detected, the address is removed from the pool for an amount of time configurable by the administrator.

52. Which command is used to build DHCP pool?

13/15 A ip dhcp pool DHCP

1/15 B ip dhcp conflict

0/15 C ip dhcp-server pool DHCP

1/15 D ip dhcp-client pool DHCP

53. What command can you enter in config mode to create DHCP pool?

14/14 **A** ip dhcp pool DHCP_pool

0/14 **B** ip dhcp exclude

0/14 **C** ip dhcp conflict logging

0/14 **D** service dhcp

54. Which command can you enter to determine the addresses that have been assigned on a DHCP Server?

0/19 **A** show ip dhcp database

1/19 **B** show ip dhcp pool

17/19 **C** show ip dhcp binding

1/19 **D** show ip dhcp server statistic

56. Which statement describes the process of dynamically assigning IP addresses by the DHCP server?

0/14 **A** Addresses are allocated after a negotiation between the server and the host to determine the length of the agreement.

0/14 **B** Addresses are permanently assigned so that the hosts uses the same address at all times.

0/14 **C** Addresses are assigned for a fixed period of time, at the end of the period, a new request for an address must be made.

14/14 **D** Addresses are leased to hosts, which periodically contact the DHCP server to renew the lease.

Gak ada yg Namanya ip permanen kecuali static

57. How does a DHCP server dynamically assign IP addresses to hosts?

11/13 **A** Addresses are permanently assigned so that the host uses the same address at all times.

0/13 **B** Addresses are assigned for a fixed period of time. At the end of the period, a new request for an address must be made, and another address is then assigned.

2/13 **C** Addresses are leased to hosts. A host will usually keep the same address by periodically contacting the DHCP server to renew the lease.

0/13 **D** Addresses are allocated after a negotiation between the server and the host to determine the length of the agreement.

58. When a DHCP server is configured, which IP address should never be assignable to hosts?

16/16 **A** network or subnetwork IP address

0/16 **B** IP address leased to the LAN

0/16 **C** IP address used by the interfaces

0/16 **D** designated IP address to the DHCP server

Ada satu lagi broadcast address

59. What is the purpose of the DHCP server?

0/19 A to provide storage for email

0/19 B to translate URLs to IP addresses

3/19 C to translate IPv4 addresses to MAC addresses

16/19 D to provide an IP configuration information to hosts

60. How is the message sent from a PC when attempts to contact the DHCPv4 server for the first time?

2/11 A Layer 3 unicast

7/11 B Layer 3 broadcast

2/11 C Layer 3 multicast

0/11 D Layer 3 anycast

61. Which mode are in PAgP?

17/17 A Auto

0/17 B Manual

0/17 C Active

0/17 D Passive

Gak ada mode manual

LACP PASSIVE ACTIVE

62. Which statement is true regarding the Port Aggregation Protocol?

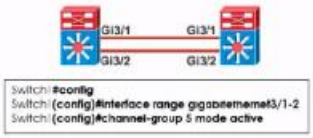
10/13 A Configuration changes made on the port-channel interface apply to all physical ports assigned to the port-channel interface.

0/13 B Configuration changes made on a physical port that is a member of a port-channel interface apply to the port-channel interface.

3/13 C Configuration changes are not permitted with Port Aggregation Protocol; instead, the standardized Link Aggregation Control Protocol should be used if configuration changes are required.

0/13 D The physical port must first be disassociated from the port-channel interface before any configuration changes can be made.

63. Refer to the exhibit. LACP has been configured on Switch1 as shown. Which is the correct command set to configure LACP on Switch2?



1/18 **A** Switch2# configure terminal
Switch2(config)# interface range gigabitethernet3/1 -2
Switch2(config-if)# channel-group 5 mode auto

14/18 **B** Switch2# configure terminal
Switch2(config)# interface range gigabitethemet3/1 -2
Switch2(config-if)# channel-group 5 mode passive

2/18 **C** Switch2# configure terminal
Switch2(config)# interface range gigabitethernet3/1 -2
Switch2(config-if)# channel-group 5 mode desirable

1/18 **D** Switch2# configure terminal
Switch2(config)# interface range gigabitethernet3/1 -2
Switch2(config-if)# channel-group 5 mode on

64. Which statement best describes implementing a Layer 3 EtherChannel?

0/18 **A** EtherChannel is a Layer 2 and not a Layer 3 feature.

10/18 **B** Implementation requires switchport mode trunk and matching parameters between switches.

8/18 **C** Implementation requires disabling switchport mode.

0/18 **D** A Layer 3 address is assigned to the channel-group interface.

B untuk Layer 2 estherchannel

65. What is the purpose of Spanning Tree Protocol in a switched LAN?

0/12 **A** To provide a mechanism for network monitoring in switched environments

1/12 **B** To prevent routing loops in networks with redundant paths

11/12 **C** To prevent switching loops in networks with redundant switched paths

0/12 **D** To manage the VLAN database across multiple switches

66. Which statement describes a spanning-tree network that has converged?

8/18 **A** All switch and bridge ports are in the forwarding state.

3/18 **B** All switch and bridge ports are assigned as either root or designated ports

7/18 **C** All switch and bridge ports are in either the forwarding or blocking state.

0/18 **D** All switch and bridge ports are either blocking or looping.

67. What does a switch do when a frame is received on an interface and the destination hardware address is unknown or not in the filter table?

1/14 **A** Forwards the switch to the first available link

0/14 **B** Drops the frame

13/14 **C** Floods the network with the frame looking for the device

0/14 **D** Sends back a message to the originating station asking for a name resolution

68. Which of the following statement is true?

1/11 **A** A switch creates a single collision domain and a single broadcast domain. A router creates a single collision domain.

4/11 **B** A switch creates separate collision domains but one broadcast domain. A router provides a separate broadcast domain.

2/11 **C** A switch creates a single collision domain and separate broadcast domains. A router provides a separate broadcast domain as well.

4/11 **D** A switch creates separate collision domains and separate broadcast domains. A router provides separate collision domains.

Switch Collison Domain → nPort, Broadcast Domain → 1

Hub Collison Domain → 1, Broadcast Domain → 1

Router Collison Domain → nPort, Broadcast Domain → nNetwork

Broadcast Domain hanya terjadi di layer 2

69. You have two switches connected together with two crossover cables for redundancy, and STP is disabled. Which of the following will happen between the switches?

5/12 **A** The routing tables on the switches will not update.

0/12 **B** The MAC forward/filter table will not update on the switch.

7/12 **C** Broadcast storms will occur on the switched network.

0/12 **D** The switches will automatically load-balance between the two links.

Routing table layer 3

70. If a switch receives a frame and the source MAC address is not in the MAC address table but the destination address is, what will the switch do with the frame?

0/19 **A** Discard it and send an error message back to the originating host

0/19 **B** Flood the network with the frame

18/19 **C** Add the source address and port to the MAC address table and forward the frame out the destination port

1/19 **D** Add the destination to the MAC address table and then forward the frame

71. The network administrator cannot connect to Switch1 over a Telnet session, although the hosts attached to Switch1 can ping the interface Fa0/0 of the router. Given the information in the graphic and assuming that the router and Switch2 are configured properly, which of the following commands should be issued on Switch1 to correct this problem?



0/11 **A** Switch1 (config)# line con0
Switch1 (config-line)# password cisco
Switch1 (config-line)#login

0/11 **B** Switch1 (config)# interface fa0/1
Switch 1(config-if)# ip address 192.168.24.3 255.255.255.0

11/11 **C** Switch1 (config)# ip default-gateway 192.168.24.1

0/11 **D** Switch1 (config)# interface fa0/1
Switch 1(config-if)# switchport mode trunk

72. Refer to the exhibit. What is the effect of the given configuration?

```
----------------------------------------------------
Switch#configuration terminal
Switch#interface VLAN 1
Switch(config-if)#ip address 192.168.2.2 255.255.255.0
Switch(config-if)#end
----------------------------------------------------
```

4/9 **A** It configures an inactive switch virtual interface.

1/9 **B** It configures an active management interface.

3/9 **C** It configures the native VLAN.

1/9 **D** It configures the default VLAN.

73. Which statement about switch access ports is true?

16/16 **A** They drop packets with 802.1Q tags.

0/16 **B** A VLAN must be assigned to an access port before it is created.

0/16 **C** They can receive traffic from more than one VLAN with no voice support

0/16 **D** By default, they carry traffic for VLAN 10.

Hanya melewatkan 1 VLAN

74. Which feature allows a device to use a switch port that is configured for half-duplex to access the network?

16/17 **A** CSMA/CD

0/17 **B** IGMP

1/17 **C** Port Security

0/17 **D** Split Horizon

75. Which option is a invalid hostname for a switch?

0/15 A 5witch-Cisco

15/15 B Switch-Cisco!

0/15 C 5witchCisc0

0/15 D SwitchCisc0

76. Which statement about unicast frame forwarding on a switch is true?

0/18 A The TCAM table stores destination MAC addresses

17/18 B If the destination MAC address is unknown, the frame is flooded to every port that is configured in the same VLAN except on the port that it was received on.

0/18 C The CAM table is used to determine whether traffic is permitted or denied on a switch

1/18 D The source address is used to determine the switch port to which a frame is forwarded

77. Two hosts are attached to a switch with the default configuration. Which statement about the configuration is true?

3/11 A IP routing must be enabled to allow the two hosts to communicate.

8/11 B The two hosts are in the same broadcast domain.

0/11 C The switch must be configured with a VLAN to allow the two hosts to communicate.

0/11 D Port security prevents the hosts from connecting to the switch.

78. Configuration of which option is required on a Cisco switch for the Cisco IP phone to work?

0/19 A PortFast on the interface

17/19 B the interface as an access port to allow the voice VLAN ID

2/19 C a voice VLAN ID in interface and global configuration mode

0/19 D Cisco Discovery Protocol in global configuration mode

79. A technician has installed SwitchB and needs to configure it for remote access from the management workstation connected SwitchA. Which set of commands is required to accomplish this task?

0/14 **A**
```
SwitchB(config)#interface FastEthernet 0/1
SwitchB(config-if)#ip address 192.168.8.252 255.255.255.0
SwitchB(config-if)#no shutdown
```

10/14 **B**
```
SwitchB(config)#ip default-gateway 192.168.8.254
SwitchB(config)#interface vlan 1
SwitchB(config-if)#ip address 192.168.8.252 255.255.255.0
SwitchB(config-if)#no shutdown
```

1/14 **C**
```
SwitchB(config)#interface vlan 1
SwitchB(config-if)#ip address 192.168.8.252 255.255.255.0
SwitchB(config-if)#ip default-gateway 192.168.8.254 255.255.255.0
SwitchB(config-if)#no shutdown
```

3/14 **D**
```
SwitchB(config)#ip default-network 192.168.8.254
SwitchB(config)#interface vlan 1
SwitchB(config-if)#ip address 192.168.8.252 255.255.255.0
SwitchB(config-if)#no shutdown
```

80. Which switching method duplicates the first six bytes of a frame before making a switching decision?

0/23 **A** fragment-free switching

22/23 **B** cut-through switching

1/23 **C** store-and-forward switching

0/23 **D** ASIC switching

81. Refer to the exhibit. Which of these statements correctly describes the state of the switch once the boot process has been completed?

1/17 **A** As FastEthernet0/12 will be the last to come up, it will not be blocked by STP.

13/17 **B** Remote access management of this switch will not be possible without configuration change.

2/17 **C** More VLANs will need to be created for this switch.

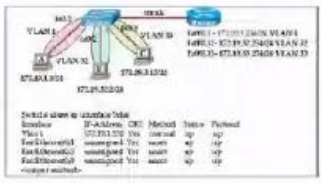
1/17 **D** The switch will need a different IOS code in order to support VLANs and STP.

Semua fa muncul bersamaan

Vlan sama semua 1001 – 1005

Vlan sama STP gak usah di ios beda

82. The network administrator normally establishes a Telnet session with the switch from host A. The administrator's attempt to establish a connect via Telnet to the switch from host B fails, but pings from host B to other two hosts are successful. What is the issue for this problem?



0/14 **A** Host B and the switch need to be in the same subnet.

14/14 **B** The switch needs an appropriate default gateway assigned.

0/14 **C** The switch interface connected to the router is down

0/14 **D** Host B need to be assigned an IP address in vlan 1.

83. What is the purpose of the OSI physical layer?

0/14 **A** controlling access to media

13/14 **B** transmitting bits across the local media

0/14 **C** performing error detection on received frames

1/14 **D** exchanging frames between nodes over physical network media

84. Which STP port role is adopted by a switch port if there is no other port with a lower cost to the root bridge?

0/17 **A** designated port

15/17 **B** root port

1/17 **C** alternate port

1/17 **D** disable port

85. What is the function of Spanning Tree Protocol in a scalable network?

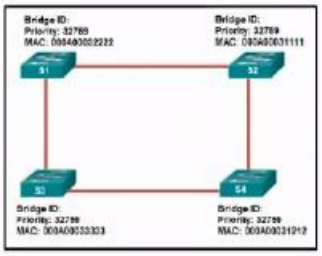
0/19 **A** It decreases the size of the failure domain to contain the impact of failures.

0/19 **B** It protects the edge of the enterprise network from malicious activity.

2/19 **C** It combines multiple switch trunk links to act as one logical link for increased bandwidth.

17/19 **D** It disables redundant paths to eliminate Layer 2 loops.

86. Refer to the exhibit. Which switch will be the root bridge after the election process is complete?



0/16 **A** S1

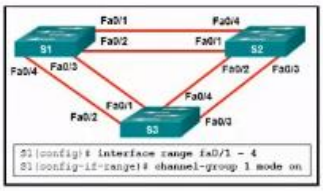
16/16 **B** S2

0/16 **C** S3

0/16 **D** S4

Liat priority → liat mac paling rendah

87. Refer to the exhibit. The administrator tried to create an EtherChannel between S1 and the other two switches via the commands that are shown, but was unsuccessful. What is the problem?



17/17 **A** Traffic cannot be sent to two different switches through the same EtherChannel link.

0/17 **B** Traffic cannot be sent to two different switches, but only to two different devices like an EtherChannel-enabled server and a switch.

0/17 **C** Traffic can only be sent to two different switches if EtherChannel is implemented on Gigabit Ethernet interfaces.

0/17 **D** Traffic can only be sent to two different switches if EtherChannel is implemented on Layer 3 switches.

88. What is requirement to be able to configure an EtherChannel between two switches?

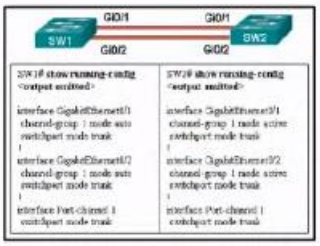
10/16 **A** All the interfaces need to work at the same speed and same duplex mode.

0/16 **B** All interfaces need to be assigned to different VLANs.

0/16 **C** Different allowed ranges of VLANs must exist on each end.

6/16 **D** The interfaces that are involved need to be contiguous on the switch.

89. Refer to the exhibit. A network administrator is configuring an EtherChannel link between two switches, SW1 and SW2. However, the EtherChannel link fails to establish. What change in configuration would correct the problem?



13/13 **A** Configure SW2 EtherChannel mode to desirable.

0/13 **B** Configure SW2 EtherChannel mode to on.

0/13 **C** Configure SW1 EtherChannel mode to on.

0/13 **D** Configure SW2 EtherChannel mode to auto.

90. A switch is configured to run STP. What term describes a non-root port that is permitted to forward traffic on the network?

1/16 **A** root port

14/16 **B** designated port

1/16 **C** alternate port

0/16 **D** disable port

C SAMA D Itu Bloking

Root mah root port sendiri

91. Which port role is assigned to the switch port that has the lowest cost to reach the root bridge?

12/12 **A** root port

0/12 **B** designated port

0/12 **C** alternate port

0/12 **D** disable port

92. A switch is configured to run STP. What term describes the switch port closest, in terms of overall cost, to the root bridge?

18/18 **A** root port

0/18 **B** designated port

0/18 **C** alternate port

0/18 **D** disable port

93. A switch is configured to run STP. What term describes the reference point for all path calculations?

13/14 **A** root bridge

1/14 **B** root port

0/14 **C** designated port

0/14 **D** alternate port

94. A switch is configured to run STP. What term describes a field that has a default value of 32,768 and is the initial deciding factor when electing a root bridge?

17/17 **A** bridge priority

0/17 **B** MAC Address

0/17 **C** extended system ID

0/17 **D** bridge ID

95. Which statement describes an EtherChannel implementation?

0/15 **A** EtherChannel operates only at Layer 2.

0/15 **B** PAgP cannot be used in conjunction with EtherChannel.

15/15 **C** A trunked port can be part of an EtherChannel bundle.

0/15 **D** EtherChannel can support up to a maximum of ten separate links.

96. Refer to the exhibit. A network administrator issued the show etherchannel summary command on the switch S1. What conclusion can be drawn?



2/15 **A** The EtherChannel is suspended.

12/15 **B** The EtherChannel is not functional.

0/15 **C** The port aggregation protocol PAgP is misconfigured.

1/15 **D** FastEthernet ports Fa0/1, Fa0/2, and Fa0/3 do not join the EtherChannel.

Ada yg sd berarti down

97. Which statement describes a characteristic of EtherChannel?

0/16 **A** It can combine up to a maximum of 4 physical links.

0/16 **B** It can bundle mixed types of 100 Mb/s and 1Gb/s Ethernet links.

16/16 **C** It is made by combining multiple physical links that are seen as one link between two switches.

0/16 **D** It consists of multiple parallel links between a switch and a router.

98. Which mode configuration setting would allow formation of an EtherChannel link between switches SW1 and SW2 without sending negotiation traffic?

11/11 **A** SW1: on
SW2: on

0/11 **B** SW1: desirable
SW2: desirable

0/11 **C** SW1: passive
SW2: active

0/11 **D** SW1: auto
SW2: auto
trunking enabled on both switches

99. When EtherChannel is configured, which mode will force an interface into a port channel without exchanging aggregation protocol packets?

0/14 **A** active

0/14 **B** auto

14/14 **C** on

0/14 **D** desirable

100. A network administrator has configured an EtherChannel between two switches that are connected via four trunk links. If the physical interface for one of the trunk links changes to a down state, what happens to the EtherChannel?

0/15 **A** Spanning Tree Protocol will transition the failed physical interface into forwarding mode.

0/15 **B** Spanning Tree Protocol will recalculate the remaining trunk links.

2/15 **C** The EtherChannel will transition to a down state.

13/15 **D** The EtherChannel will remain functional.

101. When configuring a switch for SSH access, what other command that is associated with the login local command is required to be entered on the switch?

0/19 **A** enable secret *password*

1/19 **B** password *password*

18/19 **C** username *username* secret *secret*

0/19 **D** login block-for seconds attempts number within*seconds*

102. If one end of an Ethernet connection is configured for full duplex and the other end of the connection is configured for half duplex, where would late collisions be observed?

0/13 **A** on both ends of the connection

2/13 **B** on the full-duplex end of the connection

0/13 **C** only on serial interfaces

11/13 **D** on the half-duplex end of the connection

103. Which protocol adds security to remote connections?

0/17 **A** FTP

0/17 **B** HTTP

17/17 **C** SSH

0/17 **D** Telnet

104. What is the minimum Ethernet frame size that will not be discarded by the receiver as a runt frame?

13/14 **A** 64 bytes

1/14 **B** 512 bytes

0/14 **C** 1024 bytes

0/14 **D** 1500 bytes

105. What is a characteristic of an IPv4 loopback interface?

0/11 **A** The no shutdown command is required to place this interface in an UP state.

11/11 **B** It is a logical interface internal to the router.

0/11 **C** Only one loopback interface can be enabled on a router.

0/11 **D** It is assigned to a physical port and can be connected to other devices.

106. Which impact does adding a Layer 2 switch have on a network?

0/15 **A** an increase in the number of dropped frames

14/15 **B** an increase in the size of the broadcast domain

1/15 **C** an increase in the number of network collisions

0/15 **D** an increase in the size of the collision domain

107. What information is added to the switch table from incoming frames?

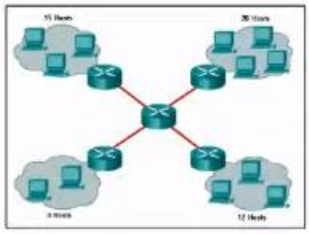
10/11 **A** source MAC address and incoming port number

1/11 **B** destination MAC address and incoming port number

0/11 **C** destination IP address and incoming port number

0/11 **D** source IP address and incoming port number

108. Refer to the exhibit. How many broadcast domains are displayed?



0/13 **A** 1

0/13 **B** 4

13/13 **C** 8

0/13 **D** 16

109. On a switch that is configured with multiple VLANs, which command will remove only VLAN 100 from the switch?

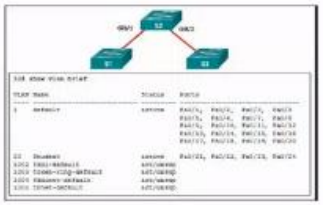
2/20 **A** Switch# delete flash:vlan.dat

0/20 **B** Switch(config-if)# no switchport access vlan 100

0/20 **C** Switch(config-if)# no switchport trunk allowed vlan 100

18/20 **D** Switch(config)# no vlan 100

110. Refer to the exhibit. A network administrator is reviewing port and VLAN assignments on switch S2 and notices that interfaces Gi0/1 and Gi0/2 are not included in the output. Why would the interfaces be missing from the output?



0/16 A There is a native VLAN mismatch between the switches

0/16 B There is no media connected to the interfaces.

0/16 C They are administratively shut down.

16/16 D They are configured as trunk interfaces.

111. What happens to switch ports after the VLAN to which they are assigned is deleted?

2/15 A The ports are disabled.

0/15 B The ports are placed in trunk mode.

0/15 C The ports are assigned to VLAN1, the default VLAN.

13/15 D The ports stop communicating with the attached devices.

112. Port Fa0/11 on a switch is assigned to VLAN 30. If the command no switchport access vlan 30 is entered on the Fa0/11 interface, what will happen?

6/16 A Port Fa0/11 will be shutdown.

0/16 B An error message would be displayed.

10/16 C Port Fa0/11 will be returned to VLAN 1.

0/16 D VLAN 30 will be deleted.

Vlan 30 dihilangkan Kembali lagi ke vlan 1

113. Which command displays the encapsulation type, the voice VLAN ID, and the access mode VLAN for the Fa0/1 interface?

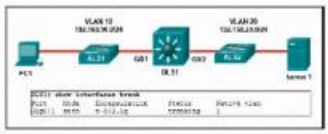
2/16 A show vlan brief

12/16 B show interfaces Fa0/1 switchport

1/16 C show mac address-table interface Fa0/1

1/16 D show interfaces trunk

114. Refer to the exhibit. PC1 is unable to communicate with server 1. The network administrator issues the show interfaces trunk command to begin troubleshooting. What conclusion can be made based on the output of this command?



13/13 A Interface G0/2 is not configured as a trunk.

0/13 B VLAN 20 has not been created.

0/13 C The encapsulation on interface G0/1 is incorrect.

0/13 D The DTP mode is incorrectly set to dynamic auto on interface G0/1.

115. Refer to the exhibit. What is the cause of the error that is displayed in the configuration of inter-VLAN routing on router CiscoVille?



5/11 **A** The gig0/0 interface does not support inter-VLAN routing.

0/11 **B** The no shutdown command has not been configured.

0/11 **C** The IP address on CiscoVille is incorrect.

6/11 **D** The encapsulation dot1Q 20 command has not been configured.

116. Refer to the exhibit. A network administrator has configured router CiscoVille with the above commands to provide inter-VLAN routing. What command will be required on a switch that is connected to the Gi0/0 interface on router CiscoVille to allow inter-VLAN routing?



0/12 **A** switchport mode access

0/12 **B** no switchport

12/12 **C** switchport mode trunk

0/12 **D** switchport mode dynamic desirable

117. A high school uses VLAN15 for the laboratory network and VLAN30 for the faculty network. What is required to enable communication between these two VLANs while using the router-on-a-stick approach?

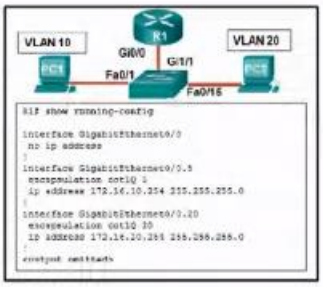
0/17 **A** A multilayer switch is needed.

0/17 **B** A router with at least two LAN interfaces is needed.

0/17 **C** Two groups of switches are needed, each with ports that are configured for one VLAN.

17/17 **D** A switch with a port that is configured as a trunk is needed when connecting to the router.

118. Refer to the exhibit. A network administrator is verifying the configuration of inter-VLAN routing. Users complain that PC2 cannot communicate with PC1. Based on the output, what is the possible cause of the problem?



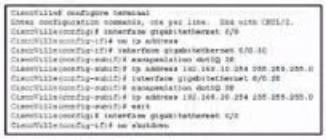
0/14 **A** Gi0/0 is not configured as a trunk port.

0/14 **B** The command interface GigabitEthernet0/0.5 was entered incorrectly.

0/14 **C** There is no IP address configured on the interface Gi0/0.

14/14 **D** The encapsulation dot1Q 5 command contains the wrong VLAN.

119. Refer to the exhibit. A network administrator has configured router CiscoVille with the above commands to provide inter-VLAN routing. What type of port will be required on a switch that is connected to Gi0/0 on router CiscoVille to allow inter-VLAN routing?



0/10 **A** routed port

0/10 **B** access port

10/10 **C** trunk port

0/10 **D** SVI

120. What is a characteristic of a routed port on a Layer 3 switch?

1/12 **A** It supports trunking.

11/12 **B** It is not assigned to a VLAN.

0/12 **C** It is commonly used as a WAN link.

0/12 **D** It cannot have an IP address assigned to it.